Executive Decision Report

Business Intelligence Capstone: Profit Growth Strategy from Sales Analytics

Prepared by: YOUR NAME | Date: DD-MM-YYYY

Executive summary: fixing discounts and scaling South can add about $43K profit a year

Key findings

- Revenue $3.36M, margin 17.0%, up 4.6% in 2025
- Orders with discounts above 20% lose $40 each on average
- South earns 35% of revenue; top 20% of customers earn 37%

Recommendations

- R1: Cap discounts at 20% with approval rules
- R2: Scale the South region playbook
- R3: Launch a top-customer retention programme

Expected impact

- Gain: $43.3K per year
- Investment: $15.1K
- Blended ROI: 187%, payback 4.2 months

Business context: growth is slow and discounting is eroding margin

Situation

- Revenue was $1.11M in 2023, $1.10M in 2024, $1.15M in 2025
- 19.5% of orders lose money
- Profit is spread unevenly across regions and customers

Objectives

- Stop margin leakage from discounting
- Grow profit in the strongest region
- Protect revenue from the top customers
- Decision: approve a three-part, $15.1K plan

Methodology: 4,000 orders over 36 months analysed in Python

Data

- Sample sales dataset, 4,000 orders
- 696 customers, 4 regions, 3 categories
- Jan 2023 to Dec 2025

Preparation

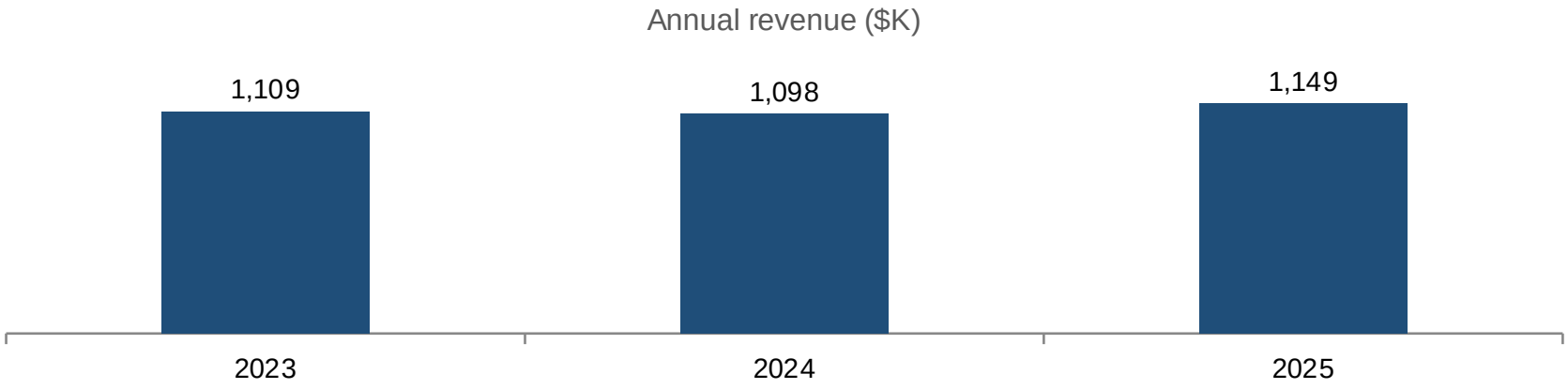
- Duplicates removed
- Dates parsed, missing values dropped
- Margin % and month fields derived

Techniques

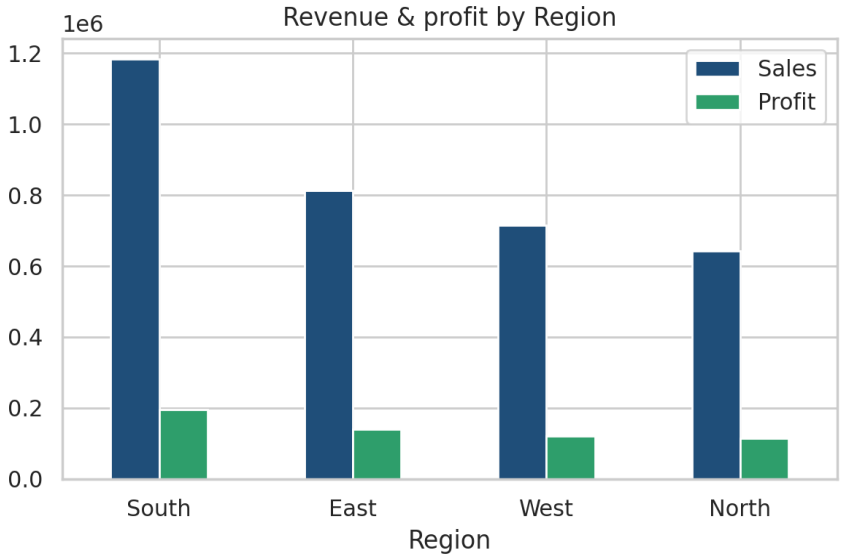
- KPI and trend analysis
- Segmentation by region, category, customer
- Three-scenario ROI modelling

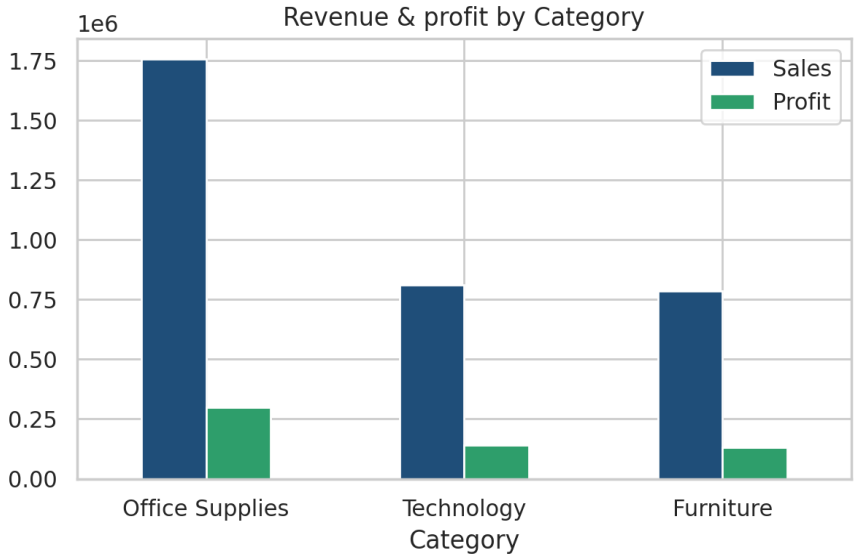
Performance snapshot: $3.36M revenue at a 17.0% margin, up 4.6% in 2025

$3.36M	$0.57M	17.0%	+4.6%
Revenue	Profit	Margin	Growth 2025



Revenue driver 1: South and Office Supplies carry the business

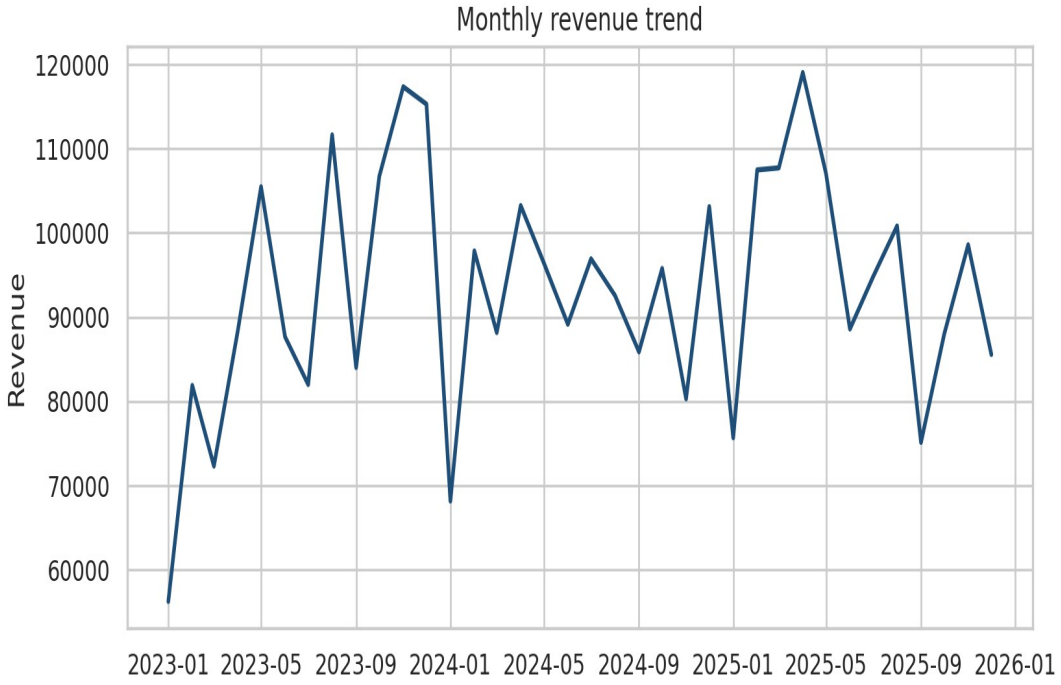




So what?

- South earns 35% of revenue and Office Supplies 52%: protect both and copy the South playbook elsewhere.

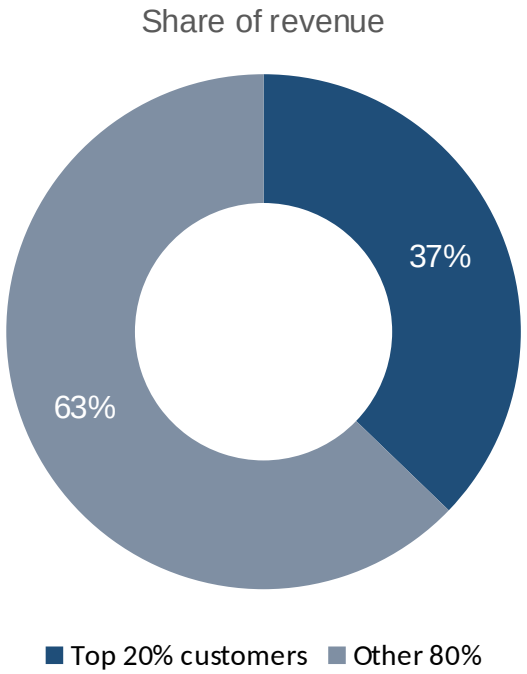
Revenue driver 2: monthly revenue is volatile with no clear upward trend



So what?

- Annual revenue moved -1.0% then +4.6%
- Growth is not coming from volume alone
- Margin and mix are the levers to pull

Revenue driver 3: the top 20% of customers generate 37% of revenue



So what?

- Concentration is moderate, not extreme
- About 139 customers drive $1.25M
- Retention here has a high payoff

Risk areas: deep discounts put about $33K of profit at risk

Margin leakage

- 823 orders with discount above 20% lose $40 each on average, $33.0K in total

Loss-making orders

- 19.5% of all orders lose money, a $34.6K drag on profit

Concentration

- South is 35% of revenue; the top 20% of customers are 37%

Data limitations

- Sample data over 36 months; costs and uplift rates are assumptions

Recommendation 1: cap discounts at 20% and require approval above it

Action

- Set a 20% hard cap in the order system
- Manager approval for exceptions
- Review discount use monthly

Rationale

- 823 orders above 20% lose $40 each on average
- Removing half of those losses is realistic
- Cheap to implement, fast to prove

Expected impact

- Annual gain: $17.1K
- Investment: $3.4K
- ROI: 408%, payback 2.4 months

Recommendation 2: scale the South region playbook to other regions

Action

- Document what South does differently
- Pilot in East, then North and West
- Add targeted sales capacity

Rationale

- South profit is $196K, the largest region
- An 8% profit lift is a modest target
- Proven model reduces execution risk

Expected impact

- Annual gain: $15.7K
- Investment: $6.7K
- ROI: 133%, payback 5.1 months

Recommendation 3: launch a retention programme for the top 20% of customers

Action

- Name an account owner per top customer
- Loyalty pricing without deep discounts
- Quarterly business reviews

Rationale

- Top 20% give 37% of revenue
- A 5% profit uplift on this group is conservative
- Retention costs less than acquisition

Expected impact

- Annual gain: $10.6K
- Investment: $5.0K
- ROI: 111%, payback 5.7 months

ROI projections: 187% expected return with payback in about four months

Annual gains and investment in $K; scenarios from ROI_Calculator.xlsx

Recommendation	Invest	Conservative	Expected	Optimistic	Payback
R1 Discount cap	3.4	10.2 (205%)	17.1 (408%)	22.2 (561%)	2.4 mo
R2 Scale South	6.7	9.4 (40%)	15.7 (133%)	20.4 (204%)	5.1 mo
R3 Retention	5.0	6.4 (27%)	10.6 (111%)	13.8 (174%)	5.7 mo
Total	**15.1**	**26.0 (72%)**	**43.3 (187%)**	**56.3 (273%)**	**4.2 mo**

Assumptions: conservative 60%, optimistic 130% of expected gain. R1 recovers 50% of losses on discounts above 20%; R2 lifts South profit 8%; R3 lifts top-customer profit 5%. Costs are 5% of revenue times 2-4%.

Implementation timeline: pilot in month 1, full rollout by month 6

Pilot (Month 0-1)	Rollout (Month 2-4)	Optimize (Month 5-6)
Activities: set discount cap; map South playbook	Activities: extend to East, North, West; assign account owners	Activities: review results, adjust caps and loyalty terms
Owner: Operations lead	Owner: Business head	Owner: Analytics team
KPI: loss-making orders %	KPI: revenue and margin per region	KPI: ROI achieved vs 187% plan

Decisions requested and next steps

Decisions needed from leadership

- Approve $15.1K budget for the three initiatives
- Name an owner for each recommendation
- Green-light the discount-cap pilot next month

Appendix & links

- GitHub: github.com/YOUR-USERNAME/executive-decision-report
- Notebook: Final_Analysis_YourName.ipynb
- ROI model: ROI_Calculator.xlsx